| INCH-POUND |

MIL-C-50785/11A (AR)
27 November 1992
SUPERSEDING
MIL-C-50785/11 (MU)
24 October 1975

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 12000011

(MODE SCALING)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50785.

REQUIREMENTS

1 Item definition. This assembly is a single-ended printed circuit board consisting of a relay driver, a -7 Vdc power supply, a mode detection circuit, a range rate detection circuit and a triangle wave generator. The assembly is used to provide mode scaling and control circuit routing within the Sight Current Generator Drawing Number 120000239 in which it is assembled.

2 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000011.

3 Government furnished property list. Not applicable.

4 Characteristics.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

4.1 Performance. The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000011.

a. Q1 and Q2 relay drivers shall control the K21 relay mounted in the sight current generator.

b. Operational amplifier AR1 shall control the -7 Vdc reference for use in the sight current generator.

c. Transistor Q3 and operational amplifier AR4 shall detect whether the system is in radar mode or manual mode.

d. Operational amplifiers AR5 and AR6 shall detect whether the range rate signal is inbound or outbound.

e. The triangle wave generator (AR2 - AR3) shall generate an 18 volt P-P triangular wave within a period of approximately 1.00 milliseconds.

4.2 Input and output levels.

4.3 Input power. Input power shall be supplied to the assembly as specified in Table I.

TABLE I - INPUT VOLTAGES

Pin	Voltages
24	+24 Vdc ± 2.0 percent
4	+15 Vdc ± 2.0 percent
14	-15 Vdc ± 2.0 percent
1 and 28	Ground

4.4 Output levels. The requirements specified in 4.5 through 4.10 shall be met when supplied with the specified voltage input.

4.5 Pin 3 output. The output shall be as specified in a and b.

a. +13 Vdc ± 15 percent between pin 3 and pin one when -15 Vdc is applied between pin 19 and pin one.

b. -13 Vdc ± 15 percent between pin 3 and pin one when +15 Vdc is applied between pin 19 and pin one.

4.6 Pin 6 output. The output shall be as specified in a and b.

a. Zero ± .01 Vdc between pin 6 and pin one when +15 Vdc is applied between pin 19 and pin one.

b. +15 Vdc ± 2 percent between pin 6 and pin one when -15 Vdc is applied between pin 19 and pin one.

Source: https://assist.dla.mil -- Downloaded: 20261-22T20:38Z
Check the source to verify that this is the current version before use.

4.7 Pin 11 output. The output shall be a triangular wave with an amplitude from -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 100 ± .11 milliseconds.

4.8 Pin 12 output. The output shall be as specified in a and b.

a. +13 Vdc ± 15 percent between pin 12 and pin one when zero Vdc is applied between pin 21 and pin one.

b. -13 Vdc ± 15 percent between pin 12 and pin one when +24.0 Vdc ± 15 percent is applied between pin 21 and pin one.

4.9 Pin 13 output. R6 shall be varied between 6.8 and 7.2 Vdc then adjusted as necessary to obtain -7.0 Vdc ± .01 percent between pin 13 and pin one. The output shall not vary more than ± .1% between pin 13 and pin 1 during temperature extremes of -40 degrees F to +160 degrees F.

4.10 Pin 20 output. The output shall be as specified in a through d.

a. +24 Vdc ± 15.0 percent between pin 20 and pin one with zero Vdc applied between pin 26 and pin one.

b. Zero ± .02 Vdc between pin 20 and pin with +15 Vdc applied between pin 26 and pin one.

c. +24 Vdc ± 15 percent between pin 20 and pin one with +24.0 Vdc ± 5.0 percent applied between pin 23 and pin one.

d. +24 Vdc ± 15 percent between pin 20 and pin one with +24.0 Vdc ± 5.0 percent applied between pin 21 and pin one.

5. Test equipment. The test equipment listed below shall be used in the performance of the applicable test methods and procedures as specified in 6.

5.1 Power supply (PS1). The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.2 Power supply (PS2). The power supply shall be -15 ± 0.1 Vdc, 100 milleamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.3 Power supply (PS3). The power supply shall be +24 Vdc ± 2.0 percent, with regulation to ± .01 percent, 10 ma or greater.

5.4 Resistors. Two resistors are required with a rating of 1100 ohms ± 5.0 percent, 2.0 watts.

5.6 Switches. Four double pole, single throw are required.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:38Z
Check the source to verify that this is the current version before use.

5.7 Digital voltmeter (DVM). The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

6 Functional test.

6.1 Test connections. Power supplies PS1, PS2 and PS3 and resistors shall be connected to the assembly under test in accordance with Figure 1 using receptacle 10545455.

6.2 Test procedure. Compliance with the requirements of 4.3 through 4.10 shall be verified as follows:

a. With switch S3 set to the -15 Vdc position, the voltage measured at pin 3 with DVM shall be +13 Vdc ± 15 percent.

b. With switch S1 set to the +15 Vdc position, the voltage measured at pin 3 with the DVM shall be -13 Vdc ± 15 percent.

c. With switch S1 set to the -15 Vdc position, the voltage measured at pin 6 with the DVM shall be +15 Vdc ± 2 percent.

d. With switch S1 set to the +15 Vdc position, the voltage measured at pin 6 with the DVM shall be zero +0.1 Vdc.

e. With the oscilloscope connected to pin 11, the voltage shall be a triangular wave with an amplitude from -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 1.00 ± 11 milliseconds.

f. With switch S4 off, the voltage at pin 12 measured with the DVM shall be +13 Vdc ± 15 percent.

g. With switch S4 on, the voltage at pin 12 measured with the DVM shall be -13 Vdc ± 15 percent.

h. With the DVM connected to pin 13, the output shall be varied between 6.8 and 7.2 Vdc by adjusting R6 and then set at a voltage of -7 Vdc ± .01 percent. This output shall not vary more than ± .1% during temperature extremes after being set at -7 Vdc ± 0.01% room ambient.

i. With switches S4, S2, and S3 set to the off position, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

j. With switch S3 set to on and S3 and S2 off, the voltage measured at pin 20 with the DVM shall be zero ± .02 Vdc.

k. With switch S2 and S3 set to on and S4 off, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

l. With switch S4 and S3 set to on and S2 off, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:38Z
Check the source to verify that this is the current version before use.

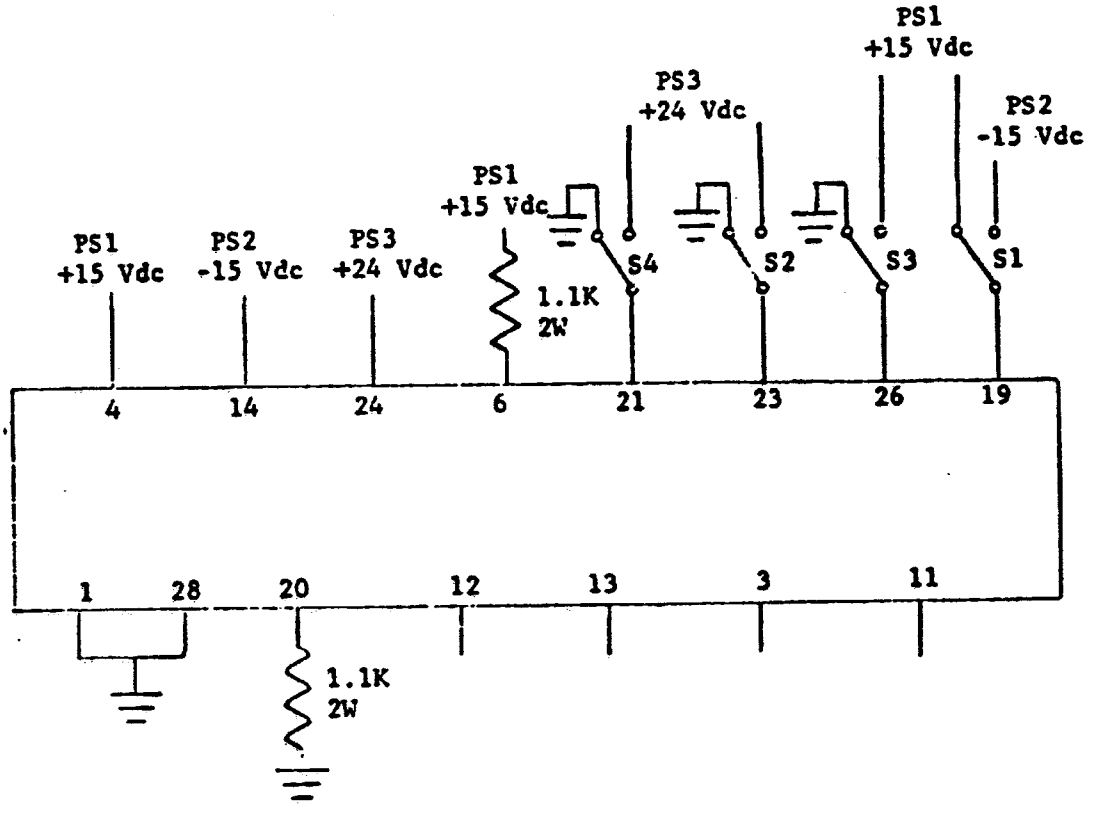


FIGURE 1. Test connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.

7.0 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing Activity:
Army - AR

Project No. 1285-A430

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:38Z
Check the source to verify that this is the current version before use.